

Functional Requirements

AIM: To identify and document functional requirements for the B&B Reservation System.

R.1 Role Selection

Description: The system shall display a home page that allows users to choose their role.

R.1.1 Select User Role

Input:

- Owner
- Customer

Output:

- Customer → Redirect to Reservation Page
- Owner → Redirect to Login Page

Processing:

- Identify selected role
- Redirect user to appropriate interface

R.2 Customer Registration

Description: The system allows customers to register before making reservations.

R.2.1 Enter Customer Details

Input:

- Name
- Address
- Phone Number

Output:

- Registration successful message

Processing:

- Validate entered data
- Store customer information in database

R.3 Room Availability Check

Description: The system allows customers to check room availability.

R.3.1 Enter Booking Dates**Input:**

- Check-in Date
- Check-out Date

Output:

- List of available rooms

Processing:

- Compare requested dates with existing reservations
- Display available rooms

R.4 Room Reservation

Description: The system allows customers to reserve rooms.

R.4.1 Enter Reservation Details**Input:**

- Customer Name
- Phone Number
- Room Number
- Check-in Date
- Check-out Date
- Agreed Price

Output:

- Reservation created successfully

Processing:

- Validate input details
- Check room availability
- Store reservation record

R.5 Payment Guarantee

Description: Reservation must be guaranteed by advance payment.

R.5.1 Enter Payment Details

Input:

- Reservation ID
- Credit Card Number
- Amount

Output:

- Payment success / failure message

Processing:

- Validate payment data
- Update reservation status

R.6 Reservation Expiry

Description: The system cancels reservations if payment is not made within the deadline.

R.6.1 Automatic Status Check

Input:

- System Date & Time

Output:

- Reservation confirmed OR cancelled

Processing:

- Check deadline

- Cancel unpaid reservations

R.7 Owner Login

Description: The system allows the owner to login securely.

R.7.1 Enter Login Credentials

Input:

- Username
- Password

Output:

- Owner dashboard

Processing:

- Validate credentials
- Authenticate owner

R.8 Expense Management

Description: The system allows the owner to record expenses.

R.8.1 Add Expense

Input:

- Description
- Amount
- Date

Output:

- Expense added successfully

Processing:

- Validate inputs
- Save expense record

R.9 Profit Report

Description: The system calculates and displays profit.

R.9.1 Generate Profit Report

Input:

- Date Range

Output:

- Total Income
- Total Expenses
- Net Profit

Processing:

- Calculate totals
- Display report

R.10 Logout

Description: The system allows users to log out securely.

Input:

- Click Logout

Output:

- Redirect to Home Page

Processing:

- End user session
- Redirect user